higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 I
1.2 J
1.3 E
1.4 G
1.5 C
1.6 H
1.7 B
1.8 A
1.9 F
1.10 D

(10 × 1) **[10]**

QUESTION 2

2.1
- It protects the weld from atmospheric contamination, i.e. prevents gases like oxygen and hydrogen from entering the weld.
- It stabilises the arc allowing for more accurate and predictable welding.
- It provides deoxidants and fluxing agents which remove impurities from the weld.
- It provides a blanket of slag which further protects the hot metal from air.
- It provides a means for adding alloying elements during welding.
- Most fluxes decrease the viscosity of the weld metal, making it flow more easily. (6)

2.2
- Water and sweat may cause electrically hazardous conditions. Shocking can be prevented by wearing insulated gloves, clothing and shoes.
- Always avoid direct contact with live electrical parts.
- Cables for arc welding should be tough and flexible. They should be designed specifically for arc welding.
- The size of the cables should be suitable for the current. Thicker cables should be used for higher current.
- Welding equipment must be checked for dust and broken electrical cables.
- Make sure that all rotating parts are in working order.
- Make sure that all flammable materials are removed from the welding area and that there is a fire extinguisher nearby for emergencies.
- It is always a good idea to keep the welding area as clean as possible. Clutter not only slows down work but also prevents one from noticing hazards which may be dangerous to one's safety. (Any 5 × 2) (10)

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2.3
- Inspecting tools and equipment is one of the most important things to do.
- When it is not correct, acknowledge this and inform the supervisor.
- Using faulty equipment places one in harm's way.
- Always stop or do not start a new job if the equipment is faulty. (4)

2.4 A – Length of weld
B – Pitch of weld
C – Size of weld
D – Arrow indicating where to weld
E – Weld all around (5 × 1) (5)
[25]

QUESTION 3

3.1
- Oxygen cylinder
- Acetylene cylinder
- Regulators
- Flashback arrestors
- Hoses
- Check valves
- Welding torch
- Nozzle/Tip (8)

3.2
- Horizontal welding
- Vertical welding
- Overhead welding (3)

3.3
- Cracking
- Surface porosity
- Incomplete root penetration
- Weld undercut
- Convexity
- Concavity
- Excessive reinforcement
- Insufficient reinforcement
- Incomplete fusion (9)

[20]

QUESTION 4

4.1
A Electrode lead
B Shielded gas inlet
C Tungsten electrode
D Shielded gas
E Solidified weld metal
F Weld pool
G Base metal
H Arc
I Gas nozzle
J Torch (10 × 1) (10)

4.2
4.2.1 Direct current electrode negative
4.2.2 Direct current electrode positive
4.2.3 Alternating current
(3 × 1) (3)

4.3
- Do not touch live electrical wires.
- Wear dry insulating gloves and body protection.
- Insulate from the work and the ground by using dry insulating mats and covers.
- Make sure the power source is disconnected before installing and servicing equipment.
- Turn off all equipment when finished.
- Do not use worn or damaged cables.
- Repair or replace all damaged parts.
- Panels and covers must be firmly in place. (Any 7 × 1) (7)

4.4
- Do not touch the workpiece with bare hands.
- Allow the workpiece to cool before handling.
- Use the proper tools or insulated welding gloves when handling the hot workpiece. (3)

4.5
- It controls the amount of shielded gas passed through the welding torch.
- Improper calibration of a flowmeter regulator will result in a too low or too high flow rate of shielding gas. (2)

[25]

QUESTION 5

5.1
- Keep the head away from the fumes.
- Work in a well-ventilated area.
- Wear an approved air respirator if ventilation is poor and cannot be corrected.
- Do not weld near degreasing, cleaning or spraying operations as the heat and rays generated from welding can react with the chemical vapours to produce highly toxic gases.
- Do not weld on coated metals because it gives off toxic fumes if welded. (5)

5.2
- It must comply with the amperage, duty cycle or indoor and outdoor use.
- It is recommended that rubber-insulated cables be selected if outdoor use is expected and PVC insulation for indoor use. (2)

5.3 An electrode wire is used to fuse metals together which creates a molten weld pool. It is melted by the arc generated when the trigger on the feeder gun is pressed. (2)

5.4 Amperage = $\frac{\text{Material thickness}}{0,0254}$ ✓

$= \frac{10}{0,0254}$ ✓✓

$= 393,7$ A✓

The correct amperage setting is 394 A.✓ (5)

5.5
- Are there protective screens in place to shield others from the arc and sparks?
- Is the work area free from cleaning agents that can react with the welding to form toxic gases?
- Is the work table secure and properly grounded?
- Is there suitable ventilation or is there a fan to clear fumes and gases?
- Is the workplace clear from any obstructions?
- Is there enough light to perform the job? (6)

[20]

TOTAL: 100

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